

Chapter 10: Determinants, the adjoint, and inverses

Theorem: If E is an elementary matrix and A is a square matrix of the same size, then $\det(EA) = (\det E)(\det A)$.

Proof:

Theorem: If A and B are $n \times n$ matrices, then $\det(AB) = \det(A) \det(B)$

Sketch of proof (if time):

Example: Compute the determinant of A , B , C , AC , and BC , where A , B , and C are the matrices

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \text{ and } C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}.$$

Confirm (if time) that $\det(AC) = \det(A) \det(C)$ and $\det(BC) = \det(B) \det(C)$.

Theorem: If A is an $n \times n$ matrix, then A is nonsingular if and only if $\det(A) \neq 0$. In this case, $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof:

Example: Suppose A, B, C, D are 3×3 matrices with $\det(A) = 4, \det(B) = -3, \det(C) = 0, \det(D) = 6$.

a.) What's $\det(A^2 B^{-1} D^T)$?

b.) Is BC an invertible matrix?

c.) If F is a 3×3 matrix with $F = P^{-1}BP$ for some invertible matrix P , what's $\det(F)$?

Example: Let A be a 4×4 matrix with $\det(A) = 3$.

(a) Describe the set of all solutions to the homogeneous system $A\mathbf{x} = \mathbf{0}$.

(b) If A is transformed to reduced row echelon form B , what is B ?

(c) Give an expression for a solution to the linear system $A\mathbf{x} = \mathbf{b}$, where

$$\mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 3 \\ 1 \end{bmatrix}.$$

(d) Can the linear system $A\mathbf{x} = \mathbf{b}$ have more than one solution? Explain.

(e) Does A^{-1} exist?

Invertibility Criteria

Let A be an $n \times n$ matrix. The following are equivalent:

1. A is invertible/nonsingular.
2. A is row equivalent to I_n .
3. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
5. The linear system $A\mathbf{x} = \mathbf{b}$ is consistent for every $n \times 1$ matrix $\mathbf{b}$.
6. A is a product of elementary matrices.
7. $\det(A) \neq 0$.

UNIT 2: VECTOR SPACES AND EIGENVALUES

Chapter 12: Introduction to Vectors

Scalars: completely described by giving their magnitude. (Mass, pressure, temperature, etc.)

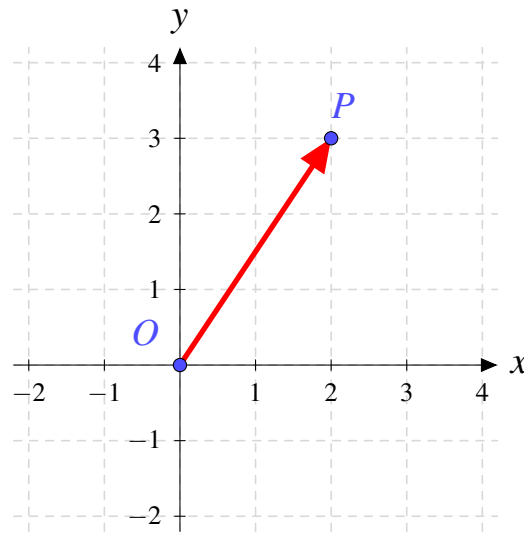
Vectors: require for their description not only a magnitude, but also a **direction**. (Velocity, acceleration, force, etc.)

Vectors

With each **point** P in the plane, we can associate an **ordered pair** of real numbers, its **coordinates**. Conversely, we can associate a point in the plane with each ordered pair of real numbers.

A **vector** can be associated with a 2×1 matrix $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$. The initial point is the **origin** and the terminal point is $P(x, y)$. The set of all vectors in the plane is denoted by $\mathbb{R}^2$, it is also called **2-space**.

Note: Your homework represents vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ as $\langle x, y \rangle$.



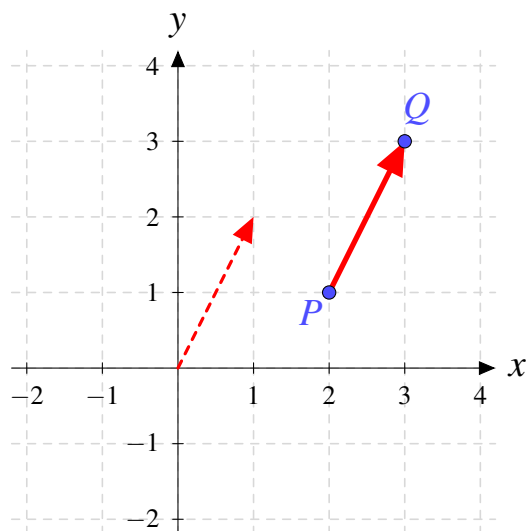
Notation: $\overrightarrow{OP}$, O is the **tail**, and P is the **head**.

A vector has a **direction**, and a **magnitude**, which is the **length** of the line segment.

The vectors $\mathbf{v} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$ are **equal** if and only if $x_1 = x_2$ and $y_1 = y_2$.

If the tail of a directed line segment is not the origin, but $P(x_1, y_1)$, and its head is $Q(x_2, y_2)$, then it can be represented by the vector

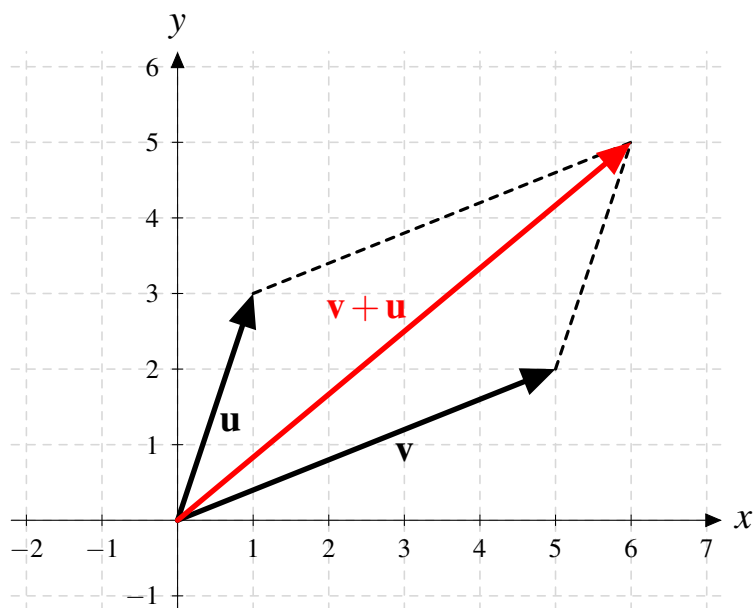
$$\overrightarrow{PQ} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix}.$$



Definition: Let $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ be two vectors. The **sum** of them is

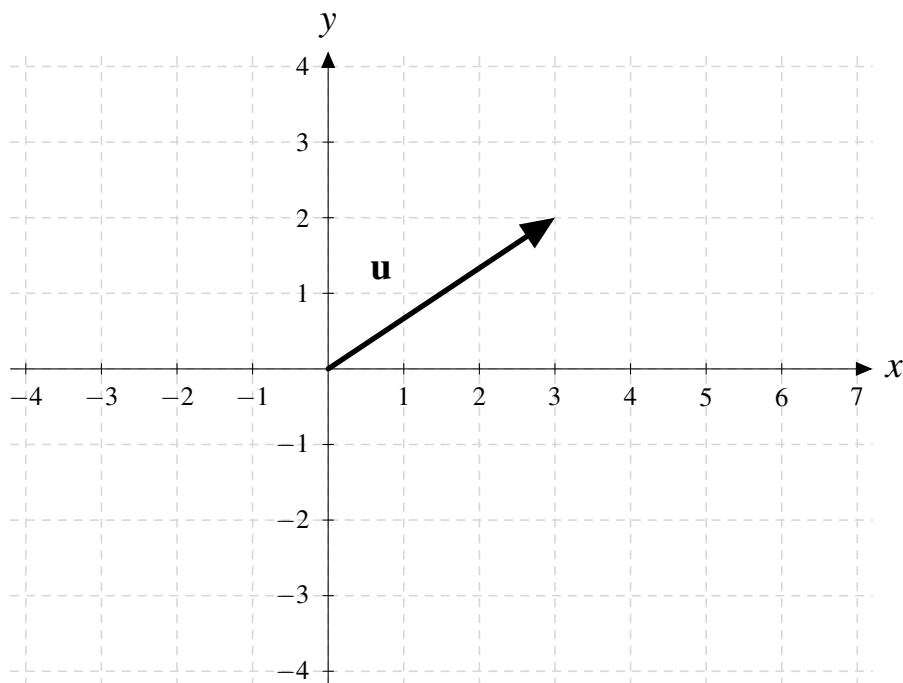
$$\mathbf{v} + \mathbf{u} = \begin{bmatrix} v_1 + u_1 \\ v_2 + u_2 \end{bmatrix}.$$

We can interpret vector addition geometrically:



Definition: If $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ is a vector, and $c \in \mathbb{R}$ is a scalar, then the **scalar multiple** $c\mathbf{v}$ is $c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}$.

Example: For $\mathbf{u}$ given below, draw $2\mathbf{u}$, $\frac{1}{2}\mathbf{u}$, and $-\mathbf{u}$.



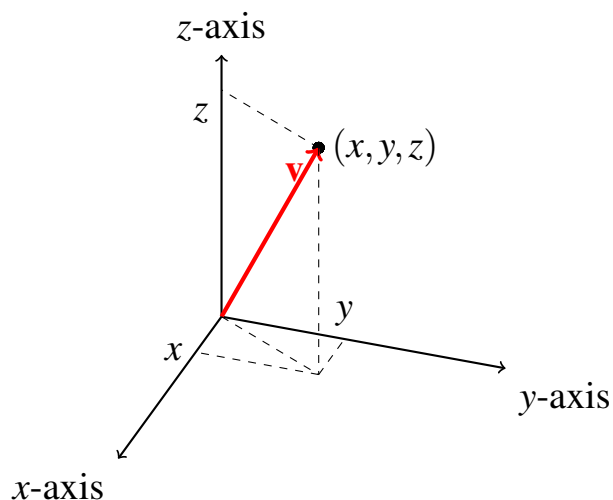
Definition: Two nonzero vectors are **parallel** if they are scalar multiples of each other.

Vectors in 3-Space and n -space

We can create vectors in higher dimensional space. In 3-space (or $\mathbb{R}^3$), we have three coordinate axes, and the origin is the point $(0,0,0)$.

With each point of the space, we can associate an ordered triple (x, y, z) of real numbers,

its **coordinates**. And we can associate a vector (or 3-vector) $\mathbf{v} = (x, y, z) = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.



Similarly to the plane, we can define addition, scalar multiplication in 3-space.

A vector in n -space (or in $\mathbb{R}^n$) is

$$\mathbf{v} = (x_1, x_2, \dots, x_n) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

where each x_i is a real number.

Similarly to $\mathbb{R}^3$, we can define addition, scalar multiplication in n -space via coordinate-wise addition/scalar multiplication.

Definition: If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are vectors in $\mathbb{R}^n$ and $c_1, c_2, \dots, c_k$ are real numbers, then an expression of the form

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k$$

is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$. The scalars $c_1, \dots, c_k$ are called **coefficients**.

Note: We've seen this definition already for general matrices.

Example: Which vectors in $\mathbb{R}^2$ are linear combinations of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$?

Example: Which vectors in $\mathbb{R}^3$ are linear combinations of $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$?

Definition: Consider $\mathbb{R}^n$, and let $\mathbf{e}_j$ denote the j th column of I_n , i.e.,

$$\mathbf{e}_j =$$

These vectors are called the **standard basis vectors** for $\mathbb{R}^n$ (this name will make sense later).

Important properties of vectors

Theorem: If $\mathbf{v}$, $\mathbf{u}$ and $\mathbf{w}$ are vectors in $\mathbb{R}^n$, and c and d are real numbers (scalars), then the following properties are valid:

1. $\mathbf{v} + \mathbf{u} = \mathbf{u} + \mathbf{v}$
2. $\mathbf{v} + (\mathbf{u} + \mathbf{w}) = (\mathbf{v} + \mathbf{u}) + \mathbf{w}$
3. $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
4. $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
5. $c(\mathbf{v} + \mathbf{u}) = c\mathbf{v} + c\mathbf{u}$
6. $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$
7. $c(d\mathbf{v}) = (cd)\mathbf{v}$
8. $1\mathbf{v} = \mathbf{v}$

Proofs of some of the properties for $\mathbb{R}^2$: